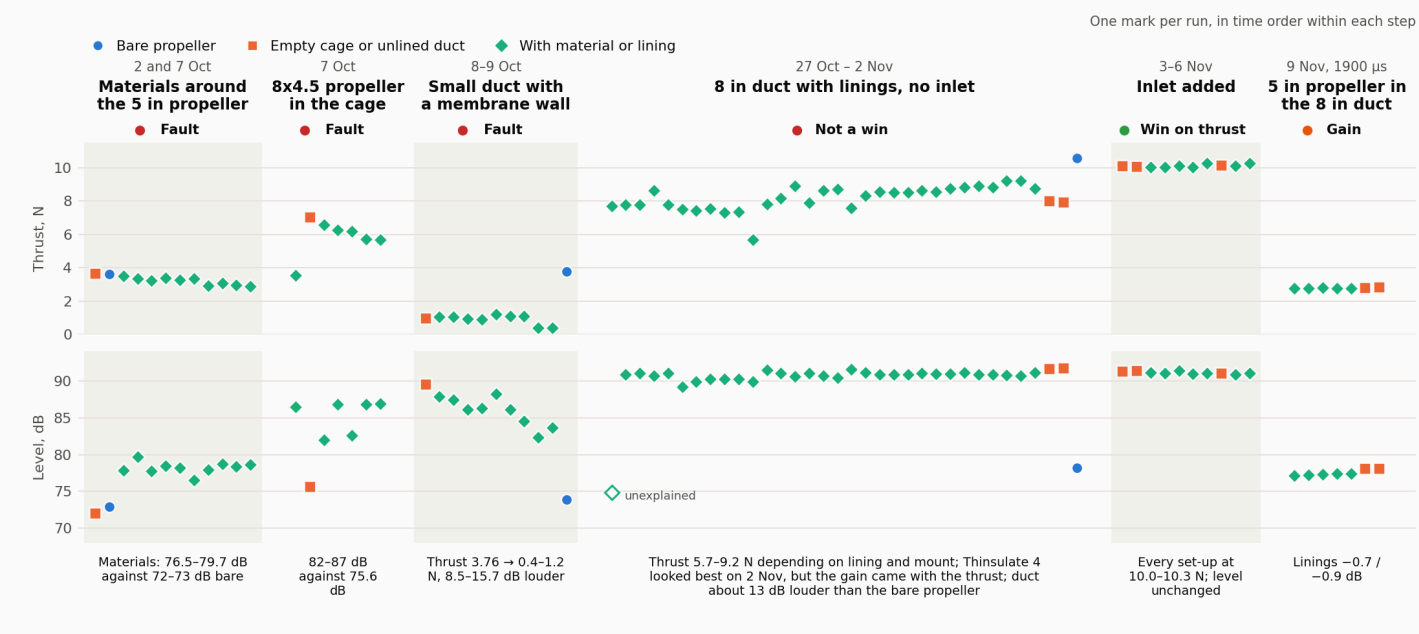
Duct and lining work, step by step

2 October – 9 November 2026. Levels are the total from 500 Hz to 20 kHz over the microphones from −54° to +90°, at 2000 µs unless stated. Each lining is compared with the unlined set-up of the same day. The reasons are Adam's account, except those marked *, which are filled in and still to be confirmed by him; the numbers come from the recordings (`CLAIMS.md` governs).



When	What we did	Why	What followed	Outcome
2 and 7 Oct	Wrapped materials around the 5 in propeller: cork, rubber, felt, felt + rubber, PU foam, Thinsulate 4 and 8, combinations. One repeat each.	First look at whether an absorbing material close to the propeller lowers the noise.*	Every material was louder than the bare propeller (76.5–79.7 dB against 72–73 dB) and lost thrust (2.9–3.5 N against 3.6 N).	**Fault**
7 Oct, afternoon	The same with the larger 8x4.5 propeller in the cage. First runs with three repeats.	To see whether the result holds for a bigger propeller, and how much one set-up scatters.*	Same result, larger: bare 7.05 N / 75.6 dB; with materials 5.7–6.6 N / 82.0–86.9 dB. Three repeats became the standard.	**Fault**
8–9 Oct	Built a small duct with an Oracover membrane wall and put ten linings behind the membrane.	A thin membrane should let the sound through to the material with little loss, so the material can work. No aerodynamic treatment was added.	The duct took most of the thrust (3.76 N bare, 0.37–1.20 N ducted) and was 8.5–15.7 dB louder than the bare propeller. The quietest linings were also the ones with the least thrust.	**Fault**
27 Oct – 2 Nov	8 in duct with the 8x4.5 propeller and linings: PETG, PU foam, Thinsulate 2–10, felt, rubber, combinations. No inlet.	A larger duct that keeps usable thrust, to compare linings in it.*	Thrust depended on the lining and its mount (5.7–9.2 N). On 2 Nov Thinsulate 4 looked best: +1.26 N, −0.95 dB against the unlined duct. But the gain came with the thrust: the plain inlet of the next step gave +2.1 N, and with it the same lining took off only 0.31 dB.	**Not a win**
3–6 Nov	Added an inlet to the duct and repeated the linings.	Thinsulate 4 seemed to work by narrowing the duct towards a better entry shape, so the entry itself was worth fixing.*	Thrust of the unlined duct rose from 8.0 to 10.1 N and every lining landed at 10.0–10.3 N. The noise did not move (91.7 against 91.4 dB), and the linings now did a third of what they did before (−0.2 / −0.3 dB).	**Win on thrust**
9 Nov	Put the 5 in propeller in the same 8 in duct, inlet on.	With the blade tips far from the wall the duct should act less on the flow and more like a cover holding the material.	The linings did more: −0.7 / −0.9 dB against 0.0 / −0.25 dB with the 8 in propeller (1900 µs). A fault appeared at 2000 µs that day, cause not found.	**Gain**

Where this leaves us

- **The inlet fixed the thrust, not the noise.** The duct with the inlet makes 10.1 N against 10.6 N for the bare propeller, and is still 13.1 dB louder than it. No lining and no inlet has touched that.
- **Once the flow was fixed, the linings mattered less.** Without the inlet Thinsulate 2 / 4 added +0.80 / +1.26 N and took off 0.60 / 0.95 dB; with it they add nothing and take off 0.22 / 0.31 dB.
- **What looked like a material gain was mostly airflow.** Of the 0.95 dB Thinsulate 4 took off without the inlet, 0.31 dB stays once the inlet fixes the flow (different days). Lining and loading changed together in every step before the inlet.

Not yet shown

- **That the membrane lets sound reach the material:** the small duct was never run with a solid wall.
- **That Thinsulate 4 matched the inlet diameter:** needs the duct's inner diameter, the fitted layer thickness and the inlet throat diameter.
- **That the duct acts less on the flow with the 5 in propeller:** no bare 5 in propeller was recorded on this motor. The lining costs the same 0.06–0.08 N with both propellers. Next run: the bare 5 in propeller on the same day as the unlined duct with the inlet.